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THE ROLE OF FREQUENCY IN PHONOLOGICAL CLITICIZATION

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Abstract

This study examines the role of frequency in phonological grammars. Specifically, I focus on the effects frequencies of verbs have on the inflected prepositions which cliticize onto them.

The study consists of a production experiment, followed by a formal analysis of the findings. In the experiment, 27 native speakers of Hebrew produced sentences with verbs that differ in their token frequency and served as potential hosts, followed by the Hebrew inflected preposition *leḥa* ‘to you’_{ms.sg.}. The frequency data were retrieved from the Hetenten corpus available on Sketch Engine (<https://www.sketchengine.eu/>). The verbs used in the experiment had either high or low token frequency. The results show that cliticization may occur regardless of the token frequency of the host. The segmental realization of the clitic, on the other hand, was found to be affected by the frequency of the host – the higher the host’s frequency, the more segmentally reduced the clitics.

The main findings of this study are twofold. First of all, frequency can be shown to be an inherent part of the phonological grammar. Secondly, cliticization has to be viewed as a gradient phenomenon - the higher the host's frequency, the more segmental reduction the clitic will undergo. This may suggest that phonology as a whole should be viewed as a gradient system, rather than a binary or privative one.

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1. Introduction

This paper presents a study of the role of token frequency in phonological cliticization of inflected prepositions in Modern Hebrew. Previous research has shown that frequency can affect various aspects of phonology, including perception (Lorenz and Tizón-Couto 2019), production (Hooper 1976b, Coetzee 2008), and historical changes (Hare and Elman 1995). In phonological production, lexical items with high token frequency tend to undergo phonological reduction. Lexical items with low token frequency, on the other hand, tend to preserve their structure and resist phonological reduction (Hooper 1976b, Coetzee 2008, Bybee 2001).

To this day, no studies on the role of frequency in cliticization have been conducted. However, it is completely reasonable to assume that, similarly to other phonological processes, frequency may play a role in phonological cliticization. When a function word cliticizes to a lexical (content) word, commonly referred to as a ‘host’, the prosodic structure of both words may be altered (Selkirk 2003). Since lexical items with low token frequency tend to preserve their structure, such hosts may block cliticization of function words to them in order to preserve their prosodic structure. Thus, a lexical word with low token frequency may resist becoming a host for a function word, resulting in a lack of cliticization and a realization of the potential clitic as a Prosodic Word rather than cliticize. Furthermore, the frequency of the host might affect the production of the clitic, such that the clitic might be more likely to undergo phonological reduction when paired with a high frequency host.

To test these assumptions, I examine the phonological behavior of Hebrew enclitic prepositions (Berman 1978, Faust and Silber-Varod 2013) in their inflected forms (Berman 1978), by conducting a production experiment.

This paper is structured as follows. Section §2 presents the relevant theoretical background, where §2.1. discusses phonological and syntactic cliticization and §2.2. provides a distinction between types of frequency and discusses the role of frequency in various aspects of phonology. Section §3 presents my research question and hypothesis. Section §4 presents the conducted experiment and section §5 discusses the findings. Section §6 discusses the prosodic structure of Hebrew cliticization. Section §7 presents a formal analysis of the findings. Finally, section §8 concludes the findings of this paper and deals with future research.

2. Theoretical background

2.1. Cliticization

A clitic is a linguistic unit that does not have phonological or syntactic independence. It attaches to either the following or the preceding lexical item. I adopt the view that, unlike an affix, which is necessarily a bound morpheme, a clitic is a bound allomorph of a morpheme that can also be realized in its full form as an unbound allomorph. The most common clitics are function words, such as prepositions, pronouns, etc. (Selkirk 2003).

Cliticization occurs at the interface between two linguistic domains: phonology and syntax. Therefore, there is a need to distinguish between syntactic cliticization and phonological cliticization.

2.1.1. Syntactic Cliticization

Syntactically, a clitic forms a constituent with another lexical item. In languages with an SVO word order, a clitic usually forms a phrase with the following lexical item (a noun or a verb) and is often located in the Spec position of that phrase (Klavans 1985). Consider the following examples. The syntactic clitics appear in bold, and the syntactic phrases are underlined.

(1a) David_{VP}[went_{PP}[P[**to**] NP[school]]]

(1b) dana_{VP}[sama_{NP}[NP[et ha-max]_{ev}] PP[**fela**] PP[al ha-ful_χan]]

Dana F-PST.put the ACC.computer her on the table

‘Dana put her computer on the table’

In (1a), the preposition ‘to’ is the head of the prepositional phrase ‘to school’. As a syntactic clitic, the preposition attaches to the following noun, which serves as its host. In (1b), the possessive pronoun *fela* is not a head of some phrase. Rather, it is a complement of the preceding noun, to which it cliticizes syntactically, forming a noun phrase. Thus, syntactic cliticization is strictly determined by phrase structure and the locations of phrase boundaries. This being said, the focus of this paper is phonological rather than syntactic cliticization.

2.1.2. Phonological cliticization

Anderson (2005) describes the phonological behavior of a clitic as having two main properties. First, there is an absence of autonomous accent, meaning that a phonological clitic does not bear stress. Second, a clitic is phonologically subordinate to another word. That is, a clitic is not an independent prosodic word, and it requires a lexical word that is a Prosodic Word to

serve as its host. Based on these observations, he provides the following definition of a phonological clitic.

(2) Phonological Clitic (Anderson 2005)

A linguistic element whose phonological form is deficient in that it lacks prosodic structure at the level of a (Prosodic) word.

Another definition is provided by Selkirk (2003), where a phonological clitic is referred to as a prosodic clitic. According to Selkirk, a prosodic clitic is “a morphosyntactic unit which is not itself a Prosodic Word”.

There are several theories which provide an account of the phonological structure of cliticization. These approaches view cliticization as a process that affects prosodic structure and are based on the prosodic hierarchy (Selkirk 1978, Selkirk 1984, Nespor and Vogel 1986).

Hayes (1989) claims that a clitic combined with a lexical word form an additional component of the prosodic hierarchy, located above the Prosodic Word unit. He calls this unit a Clitic Group and defines it as “a single content word together with all contiguous grammatical words in the same syntactic constituent” (Hayes 1989). He then presents Clitic Group formation as follows.

(3) Clitic Group Formation (Hayes 1989)

- a. Every content word (lexical category) belongs to a separate Clitic Group.
- b. Definition: the host of a Clitic Group is the content word it contains.
- c. Definition: X and Y share category membership in C if C (syntactically) dominates both X and Y.
- d. Rule: Clitic words are incorporated leftward or rightward into an adjacent Clitic Group. The group selected is the one in which the clitic shares most category memberships with the host.

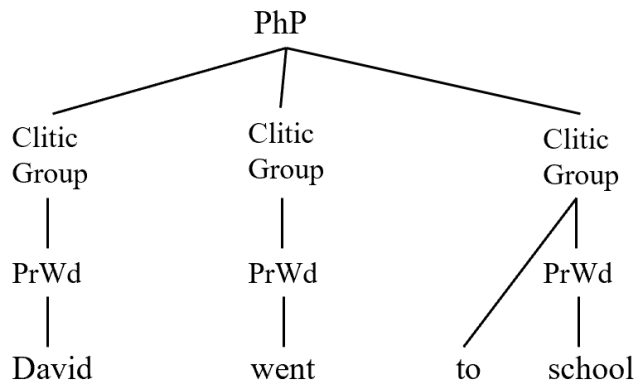
Consider the example below, repeated from (1a).

(4) David_{VP}[went_{PP}[to_{NP}[school]]]

The content words ‘David’, ‘went’, and ‘school’ are each assigned a clitic group. The clitic group that the preposition ‘to’ joins is chosen based on the number of shared category memberships with an adjacent constituent. In this case, the clitic group of ‘school’ will be

chosen over that of ‘went’, as ‘to’ and ‘school’ share more category memberships than ‘to’ and ‘went’. Therefore, the prosodic structure for the sentence will be the following.

(5) Clitic Group Structure

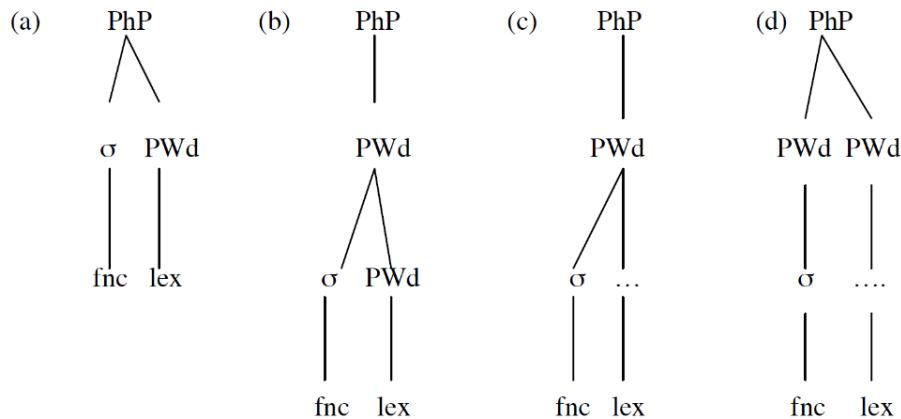


Thus, Hayes’ proposal suggests that a prosodic structure of cliticization correlates with syntactic structure. Hayes’ model, however, does not provide an account of cases where the directionality of phonological cliticization does not match the directionality of syntactic cliticization, as is the case in Hebrew, discussed in 2.1.3. below.

Another proposal accounting for the prosodic structure of cliticization is made by Selkirk (2003) and is based on the prosodic hierarchy. According to Selkirk, function words in English can appear in a strong (stressed) form or in a weak (stressless) form. Function words in their weak form are clitics, as they adhere to Andersons’ definition of a phonological clitic. A function word in its strong form is not a clitic, but rather an independent Prosodic Word, as it bears stress.

Selkirk presents four possible prosodic structures for the configuration [fnc lex], where ‘fnc’ stands for a function word and ‘lex’ for a lexical word. The models, presented in figure (6) below, show a function word preceding a lexical word, but the mirror image [lex fnc] is, of course, possible.

(6) Cliticization models



Models (a) – (c) correspond to configurations where the function word appears in its weak form and acts as a clitic. Model (d) represents the configuration [fnc lex], where the function word is in its strong form (i.e., a Prosodic Word) and does not constitute a clitic. In model (a), the clitic attaches at the Phrase level and is not part of the Prosodic Word. Selkirk (2003) refers to the clitic in this structure as a ‘free clitic’. In models (b) and (c), the clitic and the lexical word form a Prosodic Word. According to Selkirk, the clitic in (b) is an ‘internal clitic’, while the clitic in (c) is an ‘affixal clitic’. In (b), both the lexical word and the clitic are at the left edge of a prosodic word, whereas in (c), while the clitic is at the left edge of a prosodic word, the lexical word is not. Therefore, the lexical word in (b) might undergo phonological processes which occur at the left edge of a prosodic word, while the lexical word in (c) will not undergo such processes.

Selkirk proposes to view these prosodic structures through the framework of Optimality Theory (hereafter OT, Prince and Smolensky 1991, 1993). According to Selkirk, there is a set of violable constraints on prosodic structure. The relative ranking of these constraints determines the prosodic structure of cliticization in a given language.

The prosodic structure in (b) differs from that in (c) in that the clitic does not enter the Prosodic Word, but rather it forms an additional level of PrWd with its host, violating the Nonrecursivity constraint on prosodic domination (Selkirk 2003), which states that “No C_i dominates C_j , $i=j$ ” (Selkirk 2003). In other words, a Prosodic Word cannot dominate a Prosodic Word. The structure in (a) violates the Exhaustivity constraint on prosodic domination, which states that “No C_i dominates a constituent C_j , $j < i-1$ ” (Selkirk 2003). In other words, a syllable cannot be directly dominated by a Phonological Phrase. Thus, assuming constraints are violable, the

prosodic structure of cliticization in a given language will be determined by the constraint ranking of that language.

The evidence for the correct model for phonological cliticization in a given language comes from the phonological behavior of the configuration [fnc lex] with respect to processes affecting word boundaries, as presented in Selkirk (2003). The structure of cliticization of Hebrew inflected prepositions has not yet been determined. I propose that the corresponding structure is (6b), the recursive structure. Evidence for this claim is presented in section 6 of this paper.

2.1.3. Interaction between syntactic and phonological cliticization

The directionality of phonological cliticization does not always match the directionality of syntactic cliticization. In some cases, a clitic can attach syntactically to the following lexical word, but attach phonologically to the preceding lexical word, or vice versa. For example, Faust and Silber-Varod (2013) claim that Hebrew function words cliticize syntactically to the following lexical word, but cliticize phonologically to the preceding lexical word. Using Continuous Elongation boundaries, they test the directionality of phonological cliticization of Hebrew function words. They show that such boundaries can be found in natural speech after a clitic, but not before it. Faust and Silber-Varod claim that such a boundary cannot separate between a clitic and its host. Therefore, since a Hebrew clitic can be followed but not preceded by a Continuous Elongation boundary, it must be an enclitic. In the following example, *l-a* cliticizes syntactically to the following noun, while prosodically it attaches to the preceding verb.

- (7) *telex* *l-a* *noixolog* ?
 2SG.M-FUT.go to-the neurologist ?

To account for the interaction between syntactic and phonological (prosodic) cliticization, Selkirk (2011) proposes the Match Theory of syntactic-prosodic constituency correspondence. According to Match theory, there is a set of universal Match constraints that “calls for the constituent structures of syntax and phonology to correspond” and “predicts a strong tendency for phonological domains to mirror syntactic constituents” (Selkirk 2011). Thus, Match theory attempts to account for the correspondence between phonological and syntactic constituents. However, Match theory is unable to explain the differences in directionality between phonological and syntactic cliticization in certain languages, as function words are skipped over during the syntax-phonology mapping by Match constraints. According to Selkirk, a

function word may prosodically cliticize to a word that is not part of the same syntactic phrase as the function word. Match theory does not provide an account for such cases.

Another proposal regarding the differences between prosodic and syntactic cliticization is made by Klavans (1985). Klavans claims that prosodic cliticization is independent of syntax and proposes a three-parameter system for cliticization. The three parameters are Dominance, Precedence, and Phonological liaison. Dominance determines whether a clitic attaches to the initial or to the final constituent of a specific phrase. Precedence determines whether a clitic occurs before or after the host chosen by the first parameter. Both Dominance and Precedence refer to syntactic cliticization. The third parameter is phonological and determines whether the clitic is an enclitic or a proclitic and is the property of the clitic itself.

All three parameters are binary, thereby creating 8 possible configurations of cliticization. All 8 configurations are attested in languages. In 4 out of 8 configurations, the clitic is part of one syntactic constituent, while it attaches prosodically to a different constituent. Klavans refers to such clitics as having “dual citizenship” (Klavans 1985). That is, a clitic with dual citizenship has two separate hosts: a syntactic host and a phonological host.

Furthermore, Silber-Varod (2013) presents 4 models of prosody-syntax interface, showing that there is no necessary correlation between the prosodic structure of an utterance and its syntactic structure. She, thus, coins the ‘prosody first approach’ (Silber-Varod 2013), according to which the prosodic structure of a given utterance is to be studied independently of its syntactic structure.

It has been shown that phonological cliticization does not necessarily correspond with syntactic cliticization. This paper focuses on phonological cliticization.

2.1.4. Non-phonological factors in cliticization

As presented above, a function word can be realized either as a clitic or as a Prosodic Word. There are several factors that determine which realization is implemented. One such factor is whether the function word appears in focus (Selkirk 2003). Cliticization is also “constrained by pragmatic factors, in particular, degree of accessibility” (Cohen 2003). Cohen shows that this is true for pronominal possessives and proposes that other factors, such as Recoverability (Chafe 1976), may account for the cliticization of other functional elements (e.g., auxiliaries, negative particles, etc.).

Another proposal made by Ganani (2022) suggests that factors like functionality and lexical status play a role in determining whether a lexical item is realized as a clitic or as a Prosodic Word. Ganani analyzed the realizations of the Hebrew verb *tsaḥiḥ* ‘needs’ and observed their reduction patterns. She concludes that *tsaḥiḥ*, which has a lexical status of a verb, can function as an auxiliary. In its realizations as an auxiliary, its lexical status changes to that of a function word and it tends to be realized as a clitic, exhibiting phonological reduction.

This paper examines an additional potential factor that may play a role in phonological cliticization, namely, the token frequency of the potential host.

2.1.5. Cliticization of Hebrew prepositions

As presented above, prepositions in Hebrew cliticize syntactically to the following lexical item, forming a Prepositional Phrase (see figure 7). Some Hebrew monosyllabic prepositions form an orthographic contraction with the following noun, i.e., they are orthographically conjoined with the following noun. Such prepositions are *le-* ‘to’, *be-* ‘in/at’, *me-* ‘from’, and *ke-* ‘as’. Despite their syntactic and orthographic behavior that suggests that these prepositions are proclitics, these items are commonly referred to as phonological enclitics (Berman 1978).

One study supporting the claim that Hebrew monosyllabic prepositions are enclitics is that by Faust and Silber-Varod (2013). In their study, they analyzed naturally produced sentences which include either of the prepositions above, namely, *le-*, *be-*, *me-*, or *ke-*. They show that there is commonly a pause between the preposition and the following lexical word, while there is no pause between the preposition and the preceding lexical word. They explain this finding by assuming a prosodic boundary, referred to as Continuous Elongation boundary, after the preposition. Thus, Faust and Silber-Varod (2013) show that Hebrew monosyllabic prepositions form a prosodic unit with the preceding lexical item, supporting the claim that these prepositions are phonologically enclitics but syntactically proclitics.

The Hebrew enclitics are inflected when followed by a personal pronoun (Berman 1978). Thus, *be-* ‘in’, combined with a second person masculine singular pronoun *ata* forms a single unit *bexa* ‘in you’. Similarly, *le-* ‘to’, combined with a second person masculine singular pronoun *ata* forms a single unit *lexa* ‘to you’. I will refer to these forms as inflected prepositions.

2.2. Frequency

In recent decades a growing body of research on the influence of frequency on phonology has emerged (Coetzee 2008, Edwards et. al. 2015, Cohen and Ben-David 2016, among others).

When discussing the role of frequency in phonology, several distinctions must be made. First, there is a distinction between lexical frequency, i.e., the frequency of a given lexical item or lexical class, and phoneme frequency, i.e., the frequency of a given phoneme or of a natural class of phonemes in a language. Furthermore, within each category there is a distinction between ‘type frequency’ and ‘token frequency’. As defined by Bybee 2001, token frequency is “the frequency of occurrence of a unit, usually a word, in running text”, while type frequency “refers to the dictionary frequency of a particular pattern (e.g., a stress pattern, an affix, or a consonant cluster)”. That is, the term token frequency refers to how often a particular word (or another unit) occurs in speech. Type frequency, on the other hand, refers to the size of a group of words demonstrating a particular pattern (e.g., a group of words with an antepenultimate stress in Hebrew). This chapter presents a review of studies on the role of frequency in various aspects of phonology.

2.2.1. The role of frequency in production

Research on the role of frequency in phonological production shows that token frequency affects production. Specifically, frequency was found to interact with the likelihood of phonological reduction. That is, words with high token frequency are more likely to undergo phonological reduction than phonologically comparable words with lower token frequency. Coetzee (2008) discusses this phenomenon by viewing token frequency as an important factor in production of phonological variation. He does so by observing the phenomenon of final *-t/d* deletion in production of consonant clusters in American English. In English, a word final *-t/d* is sometimes deleted when preceded by another consonant (e.g., *mist* [mɪst]~[mɪs], *just* [dʒʌst]~[dʒʌs], etc.), exhibiting phonological variation. This phenomenon has been widely discussed in the last few decades (see Bybee 2002, Hooper 1976b, Labov 1989, among others). Coetzee (2008) claims that there is a correlation between token frequency and the likelihood of *-t/d* deletion. That is, final *-t/d* in words with higher token frequency are more susceptible to deletion, exhibiting higher rates of variation. Final *-t/d* in words with low token frequency, like *yeast*, *feast*, etc., on the other hand, tend to resist deletion in production. He thereby shows that token frequency plays a role in production of phonological reduction, causing phonological variation.

Another case of phonological reduction affected by token frequency is that of schwa deletion in American English reported in Zwicky (1972) and analyzed in Hooper (1976b). In American English, a schwa can be deleted if it is preceded by a stressed syllable and followed by a non-

word-final /l/, /ɪ/, or /n/. The likelihood of deletion in each of the three environments is different. However, it was shown that in the same environment the likelihood of schwa deletion depends on the token frequency of the word. That is, words with high token frequency are more likely to exhibit schwa reduction than words with low token frequency. For example, the word *memory* has high token frequency in English and is commonly realized without a schwa [memɹi]. The word *mammary*, on the other hand, has low token frequency and is commonly realized with a schwa [mæməɹi]. Thus, despite the segmental similarity, the two words are realized differently, due to differences in token frequency.

2.2.2. The role of frequency in perception

Token frequency is claimed to play a role not only in phonological production, but in perception as well. Lorenz and Tizón-Couto (2019) show that token frequency has a role in the perception of phonological reduction in American English. They conducted a word recognition experiment where participants were presented with sequences of a verb followed by ‘to’ and a verb in an infinitive form (V-to-V_{inf}) in frame sentences. The function word ‘to’ appeared in two conditions: full [tʊ] and reduced [ɹə]. The verbs preceding ‘to’ differed in their frequency. The participants reacted to the presence or absence of ‘to’ while both their accuracy and reaction times were measured.

It was found that the frequency of the preceding verb affects accuracy in recognizing the reduced variant of ‘to’. Verbs with mid-high token frequency led to high accuracy rates, while verbs with low token frequency and verbs with very high token frequency led to lower accuracy rates. Similarly, when the reduced form [ɹə] was preceded by a verb with mid-high token frequency the reaction times were significantly lower than when [ɹə] was preceded by a verb with low token frequency or a verb with very high token frequency. Thus, the study shows that token frequency has a role in perception, while the correlation is not straightforward. Rather, the mid-high range of token frequencies was found to aid perception, leading to higher accuracy rates and lower reaction times. Both low token frequency and high token frequency were shown to hinder perception, leading to lower accuracy rates and higher reaction times.

The authors explain this result by proposing that verbs with very high token frequency are processed together with the reduced [ɹə] as a single unit. They refer to this phenomenon as ‘chunking’. That is, a verb with very high frequency and the following function word are processed as a ‘chunk’. Therefore, the reaction times to the presence of ‘to’ and accuracy rates were lower in these cases, as the listeners had to extract the preposition from the ‘chunk’.

2.2.3. The role of frequency in acquisition of phonology

Research on phonological acquisition suggests that token frequency of phonemes has an effect on the order of acquisition of prosodic positions, as well as on substitution patterns. For example, in languages where /k/ has higher token frequency than /t/, /k/ is acquired earlier by children, and substitutions of /t/→/k/ take place during the acquisition stage (Yoneyama, Beckman and Edwards 2003, So & Dodd 1995, among others), despite /t/ being universally less marked than /k/ (Paradis & Prunet 1991, Steriade 1995). These findings suggest that phoneme frequency plays a role in phonological acquisition.

To examine the role of phoneme frequency in phonological acquisition, Cohen and Ben-David (2016) investigate the order of acquisition of the Hebrew rhotic /ʁ/ in various prosodic positions (i.e., in onsets, in codas, intervocalically, and in consonant clusters). The Hebrew /ʁ/ is a highly marked segment but has a high phoneme frequency in Hebrew. They show that the order of acquisition of /ʁ/ in different prosodic positions does not strictly correlate to the phonemic frequency of /ʁ/ in these positions. Rather, it correlates with the degree of phonological variation in the production of the rhotic in various positions.

The Hebrew rhotic is most commonly realized as a dorsal voiced approximant [ʁ] (Boložky 2009), but can also surface as a dorsal plosive, trill, flap, or fricative (Cohen 2013b; Cohen, Savu and Laks 2015). The degree of allophonic variation in production of /ʁ/ varies, with some positions being more prone to allophonic variation than others. Cohen and Ben-David (2016) show that /ʁ/ is first acquired by children in the prosodic position where it exhibits the least amount of allophonic variation. The order of acquisition of /ʁ/ in subsequent positions is in correlation with the amount of allophonic variation it exhibits in these positions. However, this order does not correspond to the underlying phoneme frequency of the rhotic in these positions. This finding does not provide direct evidence for the role of frequency in phoneme acquisition. Thus, some studies show the connection between frequency and acquisition order (Yoneyama et al. 2003), while others do not necessarily support frequency-based accounts (Cohen&Ben-David 2016).

Nevertheless, this finding does not contradict the claim that frequency affects phoneme acquisition patterns. The acquisition pattern can be viewed as an outcome of positional phoneme frequency in the child's perception. That is, the more frequently the rhotic is realized as an approximant in a certain prosodic position, the earlier it is acquired by a child in this

position. Thus, one can view acquisition patterns of /ʔ/ as being in correlation with type (i.e., position) and token (i.e., phoneme) frequency combined.

2.2.4. The role of frequency in language change

Token frequency has been shown to play a role in historical change, as in the case of English verbs (Hare and Elman 1995). Verbs in Old English were divided into several classes, based on the inflectional patterns of their past tense forms. The first class was a class of ‘weak verbs’, i.e., regular verbs. The remaining classes were classes of ‘strong verbs’, i.e., irregular verbs. In addition, there was a correlation between verb class and frequency. While the class of weak verbs had the most members, thus having high type frequency, its individual members had low token frequencies. For classes of strong verbs, on the other hand, the situation was reversed: each class of verbs had a low type frequency, but some individual verbs in these classes had high token frequencies.

English has gone through centuries of morpho-phonological changes from Old English, through Middle and Early-Modern English, to Modern English spoken today. In the system of verbs, the historical changes were, at least partially, determined by both type and token frequency. The strong verbs with high token frequency (e.g., went, stood, drank, etc.) survived the historical morphological changes and retained their past tense forms. As the class of weak verbs, where past tense was achieved by suffixation, had the highest type frequency, this inflection pattern survived and the class adopted even more members, including low frequency strong verbs (e.g., crept, broadcasted, etc.). Thus, type frequency of weak verbs contributed to the preservation and regularization of the morphological pattern, while token frequency contributed to the preservation of irregular forms that do not fit this pattern.

2.2.5. Summary

To summarize this section, it was shown that frequency plays a role in various aspects of phonology. Token frequency was shown to influence phonological production, such that words with high token frequency are more susceptible to phonological reduction and variation. It was shown that token frequency affects perception with tokens with mid-high frequency leading to faster and more accurate perception. Additionally, it was shown that type and token frequency combined can be seen as contributing to language acquisition, as in the case of the Hebrew rhotic. Token frequency was shown to play a role in historical changes in phonology, with words with high token frequency being less susceptible to phonological change. Type frequency was also shown to play a role in historical changes, where it allowed for the

preservation of the morphological pattern of the class with the most members (i.e., the highest type frequency).

3. Research Question and Hypotheses

The current research investigates the role of token frequency of hosts in the phonological production of clitics. It was shown that lexical items with high token frequency tend to undergo phonological reduction, like in the cases of schwa deletion (Hooper1976b) and final *-t/d* deletion (Coetzee 2008) in English. Lexical items with low token frequency, on the other hand, tend to preserve their structure and resist phonological alternation. Since cliticization has a potential to alter the prosodic structure of the host (Selkirk 2003), lexical words with low token frequency might block the cliticization of function words onto them in order to preserve their prosodic structure, given that the potential clitic can be realized as a Prosodic Word. Furthermore, the frequency of the host might affect the production of the clitic, such that the clitic might be more likely to undergo phonological reduction when paired with a high frequency host.

Thus, the main goal of this paper is to answer the following question. Does frequency play a role in cliticization? Specifically, does the token frequency of a lexical word affect the cliticization, or lack thereof, of the adjacent function word? Moreover, do clitics tend to be produced as more reduced with high frequency hosts than with low frequency hosts? To answer these questions, I examine the phonological behavior of Hebrew enclitic inflected prepositions combined with hosts that vary in their token frequencies.

The hypotheses of this study are as follows. Following Lorenz and Tizón-Couto (2019), high frequency potential hosts are expected to increase the likelihood of ‘chunking’, while low frequency potential hosts are expected to reduce the likelihood of ‘chunking’. Thus, if the token frequency of the potential host affects cliticization, hosts with low frequency will block cliticization. In this case, an inflected preposition will not tend to cliticize and will surface as a Prosodic Word that bears stress. Hosts with high token frequency, on the other hand, will not block cliticization. In this case, an inflected preposition will tend to cliticize to the host and will surface without stress. Thus, the presence or absence of stress serves as an indication of whether phonological cliticization occurred.

Furthermore, as function words generally have high token frequency, they themselves are prone to phonological reduction when they undergo cliticization. Thus, if the token frequency of the potential host affects cliticization, the degree of phonological reduction of an inflected

preposition may vary based on the token frequency of the host. That is, the higher the host frequency the more reduction the preposition will undergo. If found to be true, this will indicate that cliticization is not an absolute phenomenon, but rather a gradient one.

There are three phonological phenomena that can be viewed as phonological reduction of a segment. A segment can exhibit either a) a reduction in quantity (e.g., shortening of an unstressed vowel; b) a reduction in quality (e.g., unstressed vowels being realized as schwa); or c) deletion of the entire segment. In this paper, the examined reduction phenomenon is segmental deletion.

4. Experiment

To answer the research question and test the hypotheses presented above, a production experiment was conducted. The participants, native speakers of Hebrew, produced sentences containing a verb followed by an enclitic inflected preposition *leḥa* ‘to you’. The verbs served as potential hosts and varied in their token frequencies. The productions were then acoustically analyzed using PRAAT (Boersma and Weenink 2023), examining stress patterns as well as patterns of phonological reduction in the inflected preposition. An inflected preposition was considered a clitic if it did not bear stress and was considered reduced if one or more of its segments were deleted.

4.1. Participants

27 native Hebrew speakers, 10 male and 17 female, participated in the experiment. All participants were university students, ages 20-30. None of the participants had a personal or family history of developmental speech, language, or reading disorders.

4.2. Materials

The experiment focused on the productions of the Hebrew enclitic inflected preposition *leḥa* ‘to you’ preceded by verbs that differ in frequency. The data on the frequency of verbs in Modern Hebrew were taken from the Hetenten corpus available on Sketch Engine (<https://www.sketchengine.eu/>), which contains over 2.7 billion words. Verbs with high frequency (over 500 thousand occurrences) and verbs with low frequency (under 100 thousand occurrences) were selected. To control for syntactic structure, verbs that can be followed by *leḥa* in an indirect object position were used.

In addition, the selected verbs had to meet several phonological requirements to control for possible confounds. Verbs that end with a lateral approximant [l] were eliminated as the

following preposition begins with this consonant, which makes them indistinguishable. Similarly, verbs that ended in a rhotic were avoided if it was possible¹ as the Hebrew rhotic is acoustically similar to the Hebrew lateral. Verbs that end in a dorsal fricative [χ] were eliminated as this is the initial consonant of the reduced form [χa]. Finally, all verbs appeared in the past tense masculine singular form and were disyllabic with stress falling on the ultimate syllable. All verbs used in the experiment are presented in (8) below, accompanied by the number of occurrences in the corpus. All tokens are transcribed without glottal consonants.

(8) List of verbs and their frequencies

High-frequency verbs

katáv ‘wrote’	1,399,423
natán ‘gave’	940,335
iví ² ‘brought’	1,203,703
iḡá ³ ‘showed’	625,682

Low-frequency verbs

idpís ‘printed’	4,3487
ḡikéḡ ‘lied’	4,9656
ṡsipá ‘expected’	8,4012
baḡá ‘cried’	4,0515

Each verb was placed in a carrier sentence preceded by a subject and followed by *leḡa* and a noun phrase or a prepositional phrase. The subject, named *jósi* ‘Yossi’, was the same for all sentences. The sentences are presented in (9) below.

(9) List of sentences

High-frequency verbs	Low-frequency verbs
jósi katáv leḡa odaá ‘Yossi wrote you a message’	jósi idpís leḡa kaḡtís tisá ‘Yossi printed you a flight ticket’
jósi natán leḡa péḡaḡ ‘Yossi gave you a flower’	jósi ḡikeḡ leḡa bapaním ‘Yossi lied you in the face’
jósi iví leḡa mataná ‘Yossi brought you a present’	jósi ṡsipá leḡa mitáḡat labájit ‘Yossi expected you under the house’
jósi iḡá leḡa ṡsijúḡ ‘Yossi showed you a painting’	jósi baḡá leḡa al a-mivḡán ‘Yossi cried to you about the exam’

¹ Among the low frequency verbs, many verbs were excluded due to other requirements and one verb that ends in a rhotic was used in the experiment.

² Commonly produced as *eví*, a normatively correct form

³ Commonly produced as *eḡá*, a normatively correct form

The materials were presented in the form of a picture containing a scenario. Each picture was accompanied by the relevant verb written in Hebrew orthography. This is to ensure that the desired verb is used, as some of the low-frequency verbs have synonyms with higher frequencies. Each picture depicted two characters: Yossi, who is the agent doing the action, and Dudi, who is the recipient of the action. A picture featuring the verb *natán* ‘gave’ is presented in (10) below. All eight pictures used in the experiment are presented in appendix A.

(10) Example of a picture from the experiment



4.3. Procedure

The participants were presented with one picture at a time in rapid succession and asked to describe the action in the picture addressing Dudi. For example, for a picture depicting Yossi showing a painting to Dudi, the expected sentence is *josi iḡa leḡa tsijuḡ* ‘Yossi showed you a painting’, where ‘you’ is Dudi. The participants were instructed to produce the sentence as quickly as possible to encourage rapid speech. The pictures were presented in a randomized order. Each picture appeared ten times, not consecutively. This, too, was designed to encourage rapid and natural speech, as familiar sentences are expected to be produced more naturally and less carefully than novel sentences. Once a participant produced a sentence for a given picture, the picture was immediately replaced with the next picture.

The experiment was preceded by a brief training phase of three pictures that were not featured in the experiment. The purpose of the training was to familiarize the participants with the task.

The training phase was accompanied by additional explanations from the experimenter, if necessary.

4.4. Results

A total of 2160 sentences were produced (80 sentences per participant). 80 productions were excluded from the analysis due to mispronunciations. Of these, 66 were productions of sentences with low-frequency verbs, and 14 were productions of sentences with high-frequency verbs. For each verb, the first production of a sentence containing it was excluded from the analysis. Thus, the results presented below are based on 1864 productions of sentences. Of these, 958 are productions of sentences with a high frequency verb and 906 are productions of sentences with a low frequency verb.

The recorded sentences were acoustically analyzed using PRAAT (Boersma and Weenink 2023). For each sentence, boundaries were placed after the last segment of the verb, after the last segment of *leḥa*, and between each segment of *leḥa*. This allowed to observe both the reduction patterns of the inflected preposition (i.e., which segments in *leḥa* were not produced) and the duration of the final vowel, which, in Hebrew, corresponds with stress (Cohen et al., 2018).

4.4.1. Stress Analysis

According to Cohen et. al. (2018), vowel duration is the main parameter corresponding with stress in Hebrew, with stressed vowels in a word-final position being significantly longer than unstressed ones and having an average duration of 120ms. Intensity and pitch are viewed as complementary parameters, with stressed vowels having slightly higher intensity and pitch than unstressed ones.

In isolation, *leḥa* is realized with stress on the final syllable. The duration of the final vowel was measured to determine whether it was realized as stressed. Since clitics do not bear stress, a stressed final vowel in *leḥa* can indicate that cliticization did not occur. There were no productions of *leḥa* where the final vowel had a duration of 100ms or more. In 3 productions, the final vowel had a duration of 90-99ms. Of these, one was in a sentence where *leḥa* was preceded by a high-frequency verb, and two were in sentences where *leḥa* was preceded by a low-frequency verb. These productions are considered as possible realization of *leḥa* as a Prosodic Word that bears stress and are excluded from the reduction analysis.

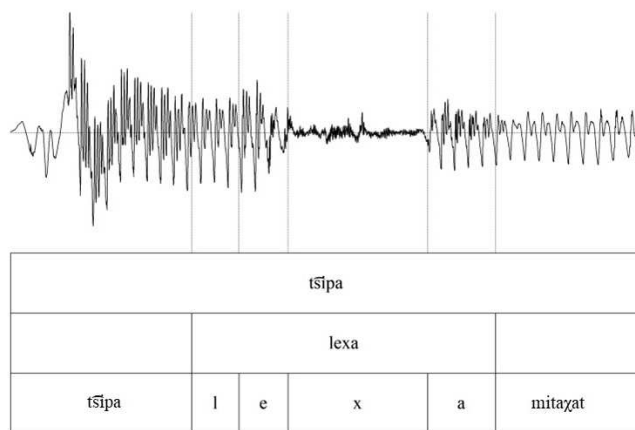
In the remaining productions of *lexa*, the duration of the final vowel did not exceed 80ms. The average duration of the final vowel was 36ms in sentences with high-frequency verbs and 40ms in sentences with low-frequency verbs. These productions are analyzed as unstressed. Thus, in all but 3 productions, *lexa* did not bear stress and was realized as a clitic.

4.4.2. Reduction Analysis

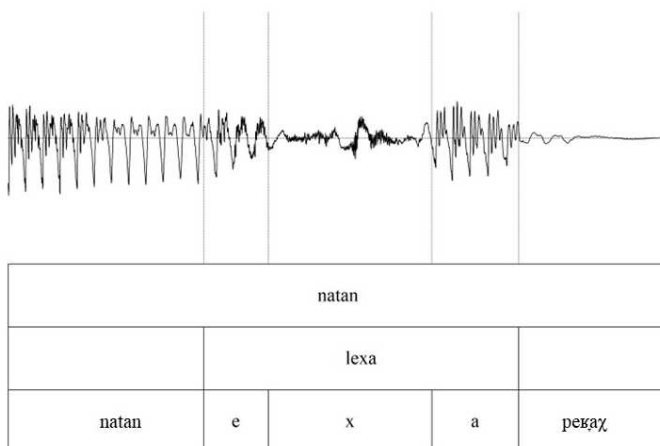
In productions where cliticization occurred, there were 4 different types of productions of *lexa*, presented in (11) below. It was produced either without reduction (11a), with reduction of [l] (11b), with reduction of [e] (11c), or with reduction of both [l] and [e] (11d).

(11) Productions of *lexa*

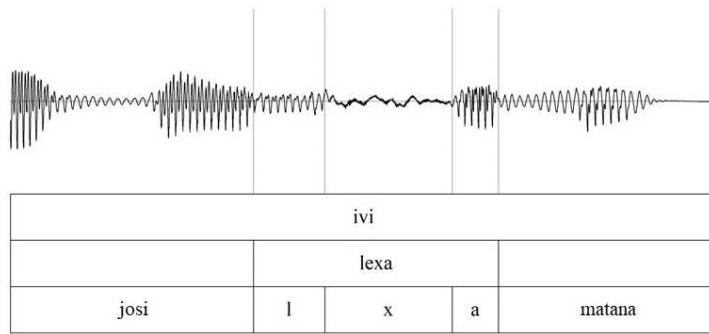
a)



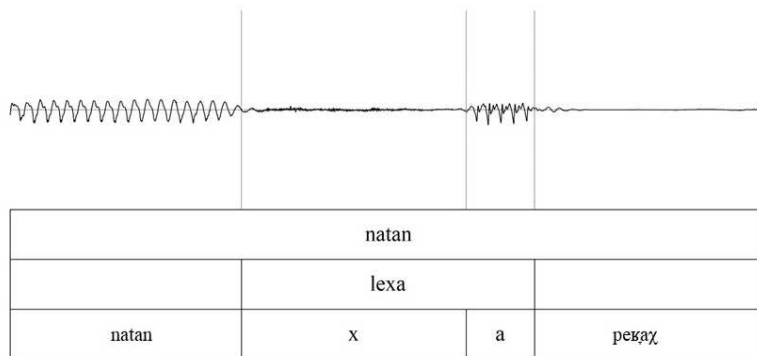
b)



c)

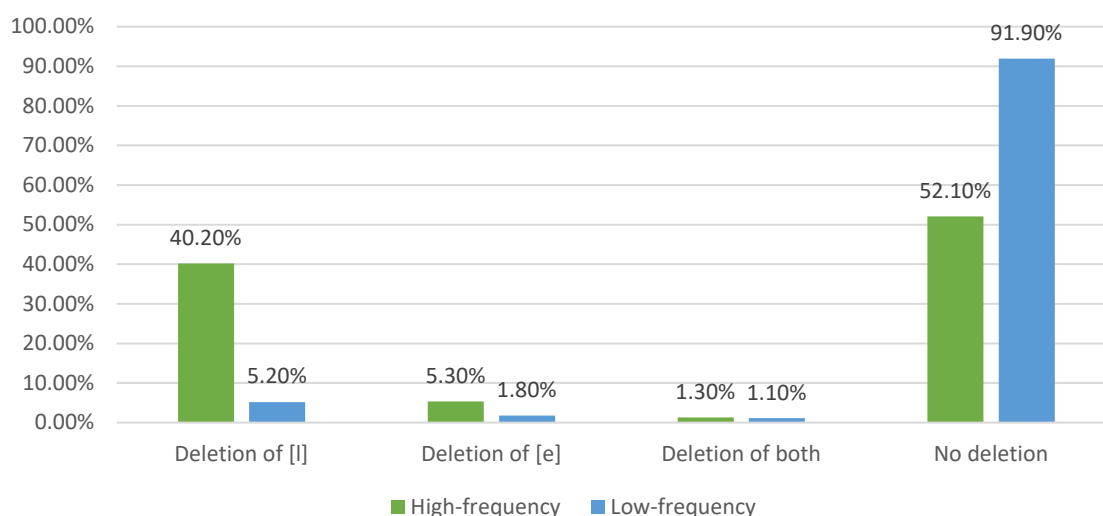


d)



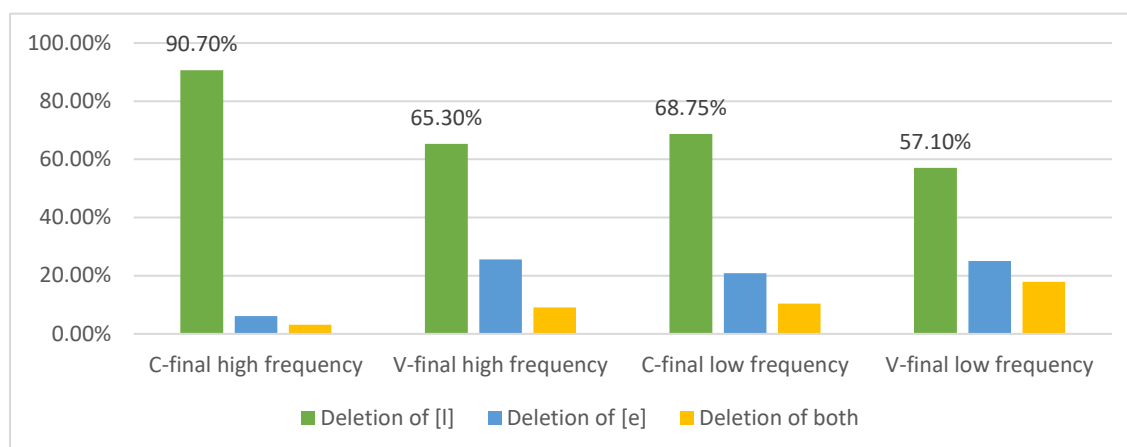
Patterns of phonological reduction (i.e., segmental deletion) were analyzed using a chi square test. It was found that reduction occurred in 47.8% of productions where *lexa* was paired with a high-frequency verb, and in 8% of productions where *lexa* was paired with a low-frequency verb. The result is statistically significant ($p < 0.05$). In productions where reduction occurred, deletion of [l] was the most common type of reduction. It occurred in 40.2% of sentences with high-frequency verbs and in 5.2% of sentences with low-frequency verbs. [e] was deleted in 5.3% of sentences with high-frequency verbs and in 1.8% of sentences with low frequency verbs. Both segments were deleted in 1.3% of sentences with high-frequency verbs and in 1.1% of sentences with low frequency verbs. Thus, deletion of [l] was the most frequent type of segmental reduction ($p < 0.05$). The results are presented in (12) below.

(12) Reduction patterns



Reduction was more common in sentences where *lexa* was preceded by a consonant-final verb than in sentences with vowel-final verbs. This is true for both high-frequency and low-frequency verbs. For sentences with high-frequency verbs, deletion of [l] was more common if the verb ended in a consonant, and deletion of [e] was more common if the verb ended in a vowel. The results are statistically significant ($p < 0.05$). For sentences with low-frequency verbs, no significant difference was found between consonant-final and vowel-final verbs ($p = 0.53$). The results are presented in (13) below.

(13) Percentage of deletion by verb structure



Furthermore, within the category of high frequency consonant final verbs, both segments were deleted more often when the preceding verb was *natan* ‘gave’. In the 11 productions where both [l] and [e] were deleted, *lexa* was preceded by *natan* in 9 productions and by *katav* ‘wrote’ in 2 productions. This suggests that even among the high frequency verbs there are differences

in frequency. Similarly, within the category of low-frequency verbs, clitics that were paired with the verb *idpis* ‘printed’ exhibited more deletion than when paired with other verbs, suggesting that there are differences in frequency among the low frequency verbs as well. Another possibility is that the nature of the final segment of the hosts affects the reduction patterns of the clitic. All consonant final verbs used in the experiment ended in different consonants (n, v, s, and ʁ). This confounding factor was impossible to avoid due to other limitations discussed in (4.2.) above.

No difference in reduction patterns was found between male and female participants. In sentences with high-frequency verbs, reduction occurred in 48% of productions of male participants and in 47.9% of productions of female speakers. In sentences with low-frequency verbs, reduction occurred in 6.9% of productions of male participants and in 8.7% of productions of female speakers. The difference is not significant ($p=0.71$).

5. Discussion

5.1. Stress

The analysis of stress patterns presented in 4.4.1. indicates that cliticization may occur regardless of the token frequency of the host. In all but 3 productions, the inflected preposition *lexa* was produced without stress. That is, *lexa* was realized as a clitic in 99.9% of productions, since a lack of an independent stress is the most prominent indicator of cliticization. This finding shows that hosts with low token frequency do not block phonological cliticization.

5.2. Phonological reduction

The analysis of patterns of phonological reduction presented in 4.4.2. indicates that reduction patterns are affected by the token frequency of the host. It was shown that when *lexa* is preceded by a high frequency verb, reduction is significantly more likely than when *lexa* is preceded by a low frequency verb. The difference in reduction patterns can only result from the token frequency of the host, as the remaining factors that can affect cliticization, such as the structure of the host, were controlled for in the conducted experiment. Given that cliticization occurs regardless of host frequency as stated in 5.1., this finding indicates that frequency is part of the phonological grammar and must be accounted for by formal theories of phonology.

It was observed that when reduction occurs, the deleted segment is most commonly [l]. The vowel [e] was also subject to deletion in some productions. The segments [x] and [a], however,

were never deleted. This begs the question of why some segments of a clitic can be deleted but others cannot. Furthermore, why does reduction occur in the first place?

The segments that are subject to deletion are determined based on the pragmatic properties of the sentence. Ariel (1990) claims that, during discourse, an addressee is instructed to retrieve a mental representation which may be characterized by reference to the individual features associated with it. The retrieval is based on the notion of Recoverability presented in Chafe (1976), who claims that the minute some information leaves consciousness some Recoverability is required for the interpretation of a form marked given, where givenness is what is assumed to be in the addressees' consciousness.

This explains why *le* can be deleted by *χa* cannot. The inflected preposition *leχa* can be decomposed into two parts: *le*, which is preposition that means 'to/for' and signifies that the action has a recipient, and *χa*, a second person masculine singular marker, which indicates the recipient (i.e., addressee). In all sentences, the verb preceding *leχa* takes a recipient argument. Thus, the first part, *le*, is predictable by the syntactic (thematic) properties of the preceding verb, which make it 'given'. Therefore, *le* can be deleted phonologically because it is subject to Recoverability. The second part of the inflected preposition, *χa*, presents new information, the addressee, which is not 'given', and, thus, not subject to Recoverability. Therefore, *χa* cannot be deleted phonologically.

The Recoverability of *le* does not imply that both segments must be deleted, but rather that they may be deleted. Other grammatical factors, including frequency, will determine whether one of these segments will be deleted in production, both segments, or neither. The formal accounts for this phenomenon are presented in §7 below.

5.3. Implications

The main implication of these findings is that phonological frequency is an inherent part of phonological grammar. The information regarding the phonological frequency of the verb, which serves as a host for a clitic, must be stored in the grammar to account for the different productions of the clitic when paired with verbs that differ in frequency. As has been noted by Cedergren and Sankoff (1974), "aspects of performance that are found to be thoroughly systematic in an individual and throughout a community are reflections of linguistic competence". In this case, the aspect of performance is variation in the production of the enclitic (i.e., with or without segmental deletion) as a function of the phonological frequency of the host. This variation has been found to be systematic among the participants of the

experiment reported in this paper. Following Cedergren and Sankoff, this indicates that the speaker's ability to produce full vs. reduced forms of the clitic is part of their linguistic competence, i.e., their phonological grammar.

Furthermore, Coetzee (2008) claims that frequency information is stored in the lexicon. He states that since frequency is an idiosyncratic property of a lexical item and cannot be deduced from its phonological properties, it must be stored with the lexical entry of each lexical item. Formal theories of phonology, therefore, must incorporate phonological frequency into the grammar. Section 7 of this paper will examine the ability of different theories of phonology to account for frequency effects. I will compare two theoretical approaches: a linear theory of phonology (i.e., rule-based phonology) versus a parallel one (i.e., Optimality Theory).

Moreover, these findings indicate that phonology should be viewed as a gradient system rather than a binary one. Recall that there was a difference in reduction patterns within the group of high frequency verbs and within the group of low frequency verbs. This indicates that the frequency scale is gradient rather than binary. The higher the frequency of the host, the more likely is the phonological reduction in the clitic.

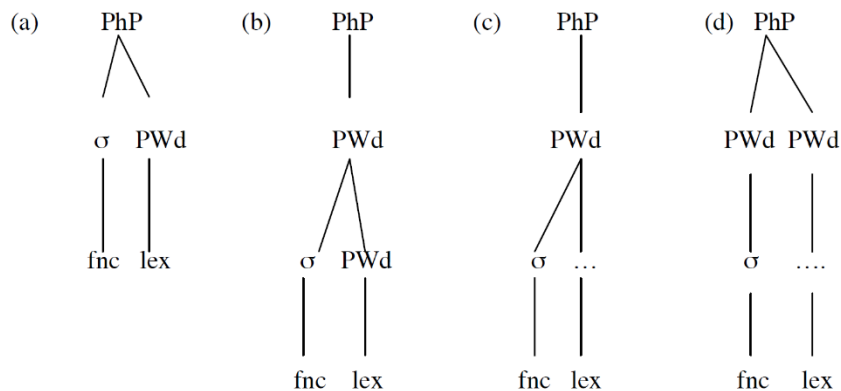
Numerous aspects of phonology are usually seen as having a specific number of values – privative (unary), binary, or multi-valent. However, it is clear that some phonological features do not behave this way. In processes of voicing assimilation, for example, the consonant undergoing the process rarely changes its voicing completely. Rather, it usually retains its original voicing for some part of the duration of its production (Mizrachi 2016). Thus, studies show that voicing assimilation tends to be gradient rather than binary (Jansen 2004, Stevens et al. 1992). While this may be seen as a sporadic post-lexical phonetic process, if the degrees can be shown to be systematically determined, it may indeed be part of the phonological grammar of the language itself. Similarly to voicing assimilation, cliticization may also be gradient. If this gradience can be shown to be systematic, this will have implications for current theories of phonology as we know it. Further research is required to provide support for this claim.

Finally, another implication of the findings is that the deletion patterns shed light on the prosodic structure of cliticization in Hebrew, presented in section 6.

6. Prosodic Structure of Cliticization

Recall the four possible models of cliticization proposed by Selkirk (2003), presented in (6) above and repeated in (14) below.

(14) Cliticization Models



The models present the function word as a proclitic but can be mirrored to present an enclitic. Faust and Silber-Varod (2013) claim that in Hebrew an enclitic preposition must join another phonological word, eliminating prosodic structures (14a) and (14d). Thus, the prosodic structure of cliticization in Hebrew is either (14b) or (14c), which differ in the presence or absence of a Prosodic Word boundary between the clitic and its host. To determine the correct prosodic structure of cliticization in Hebrew, phenomena related to PrWd boundaries should be examined. One such phenomenon is the different treatment of hiatus within a word and across words.

Within a word, hiatus is generally allowed. Sequences of vowels can be found in nouns such as *tapuáχ* ‘apple’, *mafteáχ* ‘key’, and in verbs such as *maáχ* ‘squished’, *tiém* ‘coordinated’. What is common among these examples is the historical presence of a guttural segment in their underlying representation (e.g., /mafteaħ/, /tiʔem/). Today, guttural segments are not pronounced by most speakers (Faust 2023a) and their presence in the underlying representation is an issue of debate (Boložky 1978a, 2003, Matras & Schiff 2005, Enguehard & Faust 2018, Laks et al. 2016, Bassel & Berrebi 2016).

Across a word boundary, however, the situation is different. Hebrew speakers tend to avoid hiatus by means of glide insertion. Glide insertion is especially common when a vowel-final pronoun (e.g., *aní* ‘I’) is followed by a verb in the future tense (e.g., *elex* /ʔelex/ ‘I will go). The

first person future inflection is created by adding a prefix that consists of a glottal stop and a vowel, where the glottal stop is not pronounced by most speakers. The omission of the glottal stop results in a sequence of two vowels. To avoid hiatus in this structure, speakers may insert a glide which shares the features of the preceding vowel, resulting in surface forms such as *aní jeléχ* ‘I will go’. Glide insertion is not limited to future tense verbs and can occur with past tense verbs as well (e.g., *ú wamáχ* /hu ʔamaχ/ ‘he said’). Thus, while hiatus is allowed within a word, it is avoided across a word boundary.

The results of the experiment presented above show that reduction of the initial consonant of the clitic is less likely with vowel-final hosts. These results correspond with the findings in Faust 2023a, who examined reduction patterns in Hebrew inflected prepositions, including *leχa*. He claims that when the inflected preposition encliticizes to a vowel-final verb, hiatus hinders syncope. The fact that hiatus is allowed within a word and is avoided across a word boundary indicates that a word boundary is present between the clitic and its host. Therefore, the corresponding structure of cliticization in Hebrew is structure (14b), the recursive structure, and the clitic is an ‘internal clitic’.

7. Formal Analyses

As established above, token frequency must be incorporated into models of phonological grammar. This section examines the ability of two competing theories of phonology to predict the likelihood of a full or reduced surface form of a clitic, given the frequency of its host. The theories to be examined are Optimality Theory (OT, Prince and Smolensky 1991, 1993) and the rule-based theory, introduced by Chomsky and Halle (1968).

7.1. Optimality Theory

One of the basic assumptions in OT is that constraints are violable. It is the relative ranking of constraints in the grammar of a given language that determines the optimal output for any input. When a set of candidates is submitted for evaluation, only one candidate can be deemed optimal for the given input (underlying representation). Naturally, the classic OT model does not account for phonological variation.

Throughout the years, a few accounts of frequency-based variation within Optimality Theory have been proposed. These include the Partially Ordered Constraints theory of Kiparsky (1993) and Anttila (1997) and Lexical Distribution Function approach developed in Coetzee (2008). Accounts for phonological variation caused by frequency have been proposed in other parallel

theories as well (e.g., Coetzee & Kawahara 2013). All these, however, deal with analyzing segmental processes, such as nasal place assimilation and English final -t/d deletion. To my knowledge, no accounts for frequency-based variation in cliticization have been proposed. In this paper, I will provide such an account using the model presented in Coetzee (2008).

Coetzee bases his proposal on the model of OT with indexation (Ito & Mester 1999), where each lexical item has a lexical class affiliated with it. Faithfulness constraints are indexed to indicate the lexical class affiliated with them. The same faithfulness constraint with different indexes can be ranked above or below some markedness constraint. Thus, a faithfulness constraint F_{L1} can outrank a markedness constraint M , which, in turn, outranks F_{L2} (the same faithfulness constraint belonging to lexical class 2), creating a grammar where $F_{L1} \gg M \gg F_{L2}$.

To account for frequency-based variation in this model, Coetzee proposes that lexical class affiliations are variable. That is, a single lexical item can be assigned to different lexical classes based on the frequency of that item. Each lexical item is associated with a probabilistic distribution function determining the likelihood with which it is assigned to each of the available lexical classes. The shape of the lexical distribution function is determined by the frequency of that item. It is symmetrical for items with average frequency, right-skewed for items with high frequency, and left-skewed for items with low frequency. The distribution functions range over an area which is equally divided into lexical classes. When a lexical item is submitted to the grammar, a value is chosen randomly from the distribution, which determines the class affiliation the lexical item is assigned on this specific occasion. Thus, the optimal output in one lexical class can be different than the optimal output for the same input in another lexical class.

I propose the following analysis for the segmental reduction in clitics presented in this paper. First, some markedness constraint that causes segmental reduction in the clitic is required. I propose the constraint *MinFnc*, presented in (14) below. The relevant faithfulness constraints are the standard *MAXIMALITY* constraints, divided into *MAX-C* and *MAX-V*, defined in (15) below. Finally, some faithfulness constraint will prevent the deletion of the segments χ and a which were never deleted in the productions of the clitic. This constraint is assumed to be ranked above *MinFnc* in all lexical classes and will not be discussed in this paper.

- (15) **Minimize-Fnc (MinFnc)** – Assign a violation mark for every segment of a function word in the output.

MAX-C – Every consonant in the input must have a correspondent in the output.

MAX-V – Every vowel in the input must have a correspondent in the output.

There are four lexical classes, each corresponding to one of the observed reduction patterns in the clitic. L1 is the lexical class where no deletion occurs. In L1, both MAXIMALITY constraints must outrank MinFnc. In L2, MAX-C is ranked above MinFnc, while MAX-V is ranked below it, creating an output with vowel deletion but not consonant deletion. In L3, the ranking is reversed, creating an output with consonant deletion. Finally, in L4, both MAXIMALITY constraints are ranked below MinFnc, creating an output with deletion of both segments. The lexical classes are presented in (16) below.

(16) **L1:** MAX-C_{L1}, MAX-V_{L1} >> MinFnc

L2: MAX-C_{L2} >> MinFnc >> MAX-V_{L2}

L3: MAX-V_{L3} >> MinFnc >> MAX-C_{L3}

L4: MinFnc >> MAX-C_{L4}, MAX-V_{L4}

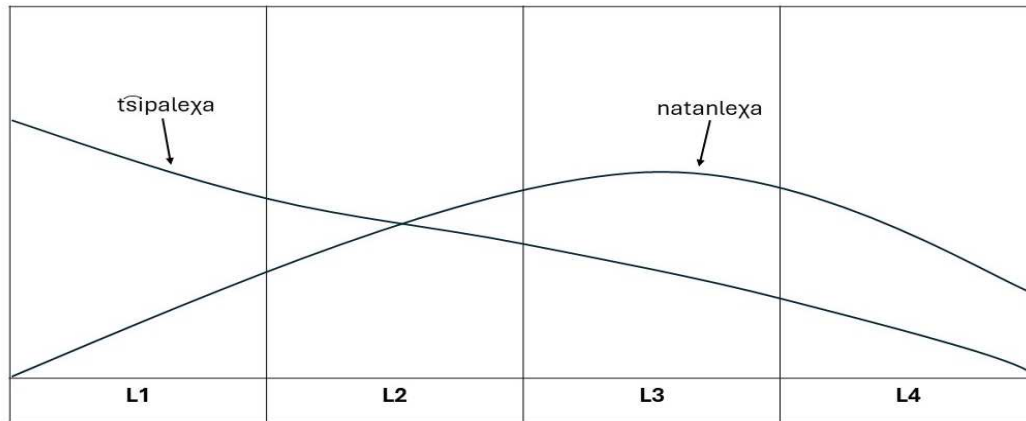
My proposal deviates from Coetzee (2008) in that the lexical classes are not necessarily gradient. In Coetzee's model, the lexical classes are gradient, where the leftmost class (L1) exhibits no deletion. Then, every lexical class exhibits deletion in more contexts than the class to the left of it. In my proposal, lexical classes can differ in the type of deletion (i.e., consonant versus vowel deletion). L2 and L3 have a reverse ranking of the MAXIMALITY constraints. Thus, L3 does not exhibit more deletion, but rather a different type of deletion. Combining the relative rankings in (16), we arrive at the following grammar.

(17) MAX-C_{L1}, MAX-V_{L1}, MAX-C_{L2}, MAX-V_{L3} >> MinFnc >> MAX-V_{L2}, MAX-C_{L3}, MAX-C_{L4}, MAX-V_{L4}

Every time that a lexical item is submitted to the grammar for evaluation, it is assigned to one specific lexical class, determining which indexed constraint will evaluate it. The lexical classes are presented linearly with L1 being the leftmost and L4 being the rightmost. The selection of the lexical class is done by selecting a random point on the lexical distribution function ranging over the four lexical classes. The shape of that function is determined by the frequency of the lexical item and is right-skewed for lexical items with high frequency and left-skewed for lexical items with low frequency. Thus, when a high frequency lexical item enters the evaluation, it can be assigned to any of the four lexical classes, but it is more likely to be

assigned to a lexical class that appears on the right (L4). Similarly, when a low frequency lexical item enters the evaluation, it is more likely to be assigned to a lexical class that appears on the left. This is illustrated in (18) below.

(18) Lexical Distribution Functions



The verb *tsipa* ‘expected’ is a low frequency verb, while the verb *natan* ‘gave’ is a high frequency verb. For the purpose of this analysis, I assume that the verb and the enclitic enter the evaluation together as a single lexical item. Note that the peak of the distribution function for *natanlexa* is located in L3 and not in L4. The results of the experiment show that even with high frequency verbs, deletion of two segments is less likely than deletion of just the initial consonant of the clitic. It is likely the case that deletion of both segments is more common with verbs that have ultra-high frequency.⁴ The fact that vowel deletion was less common than consonant deletion is also accounted for by the order of the lexical classes and the shape of the distribution function.

The tableaux in (19) illustrate how an input of *natanlexa* is evaluated under different indexations to achieve different outputs. For simplification purposes, MAX-CL_{L1} and MAX-V_{L1}, which both outrank MinFnc are collapsed into a general MAX_{L1} constraint. Similarly, MAX-CL_{L4} and MAX-V_{L4} are collapsed into MAX_{L4}.

⁴ For example, the verb *lamag* ‘said’ has the highest frequency among all Hebrew verbs according to the Hetenten corpus and can be paired with *lexa*. It was not included in the experiment as verbs ending in a rhotic were avoided.

(19) Evaluation of /natanlexa/

/natanlexa _{L1} /	MAX _{L1}	MAX- C _{L2}	MAX- V _{L3}	MinFnc	MAX- V _{L2}	MAX- C _{L3}	MAX _{L4}
a. ɛ natanlexa				****			
b. natanlχa	*!			***			
c. natanexa	*!			***			
d. natanyχa	*!*			**			

/natanlexa _{L2} /	MAX _{L1}	MAX- C _{L2}	MAX- V _{L3}	MinFnc	MAX- V _{L2}	MAX- C _{L3}	MAX _{L4}
e. natanlexa				****!			
f. ɛ natanlχa				***	*		
g. natanexa		*!		***			
h. natanyχa		*!		**	*		

/natanlexa _{L3} /	MAX _{L1}	MAX- C _{L2}	MAX- V _{L3}	MinFnc	MAX- V _{L2}	MAX- C _{L3}	MAX _{L4}
a. natanlexa				****!			
b. natanlχa			*!	***			
c. ɛ natanexa				***		*	
d. natanyχa			*!	**		*	

/natanlexa _{L4} /	MAX _{L1}	MAX- C _{L2}	MAX- V _{L3}	MinFnc	MAX- V _{L2}	MAX- C _{L3}	MAX _{L4}
e. natanlexa				***!*			
f. natanlχa				***!			*
g. natanexa				***!			*
h. ɛ natananyχa				**			**

It was shown that the Lexical Distribution Function approach developed by Coetzee (2008) within the framework of OT can successfully generate the four observed variants of the clitic. Furthermore, this approach allows to predict the likelihood of a certain surface form of the clitic based on the frequency of its host.

Note, however, that the distinction between L2 and L3 might not be necessary. These two classes differ in the type of deletion (i.e., consonant vs. vowel), but not in the number of segments deleted. It might be the case that the prosodic structure of the verb determines whether the consonant or the vowel undergoes deletion. It was found that vowel deletion in the clitic is more likely when the host ends in a vowel and consonant deletion is more likely when the host ends in a consonant. Thus, it is possible that markedness constraints that prohibit sequences of consonants and sequences of vowels are the ones to determine the deletion patterns. More data is necessary to determine whether this is the case.

7.2. Rule-based theory

In the rule-based theory, as introduced by Chomsky and Halle (1968), surface forms are achieved by ordered application of rewrite rules of the form $A \rightarrow B / C_D$. To account for variation, rewrite rules can be marked as [+optional] (Vaux 2008, among others). This notation, however, can only express the optionality of application of a given rule, but not the probability of its application. Labov (1969) noticed that the probability of application of some rules can be predicted based on the context (e.g., the preceding and following segments). He introduced the notion of weighting the components in the structural description of a rule such that each component contributes to the probability of rule application. While Labov's proposal can predict the probability of application of some rules, it does not account for the role of frequency in the application of phonological rules. To my knowledge, no accounts for the role of frequency in phonology within the rule-based theory have been proposed.

To account for frequency effects in phonological variation, I propose to incorporate frequency information into the structural description of the rule. Probability of application can be calculated based on the relative frequency in the lexicon of the word that the rule applies to and represented in form of percentage (e.g., $A \rightarrow B/C_D$ [+optional, 70%]). During a derivation, the higher the probability percentage, the more likely the rule is to apply.

Two deletion rules are required to account for the deletion patterns presented in this paper: a vowel deletion rule and a consonant deletion rule. The environment of each rule must be such that the application of one rule does not block the application of the other. The C-deletion rule can be formulated with reference to the clitic boundary alone, as presented in (19) below. The case of vowel deletion is more complex as it does not occur at any boundary. The rule, therefore, must refer to some adjacent segment. The segment to the left of the vowel is a consonant that is also subject to deletion. The segment to the right of the vowel is a consonant

that is not subject to deletion. However, in some inflected prepositions, the first vowel is followed by a vowel and not a consonant⁵. Thus, the vowel deletion rule should be formulated with reference to the left context, as presented in (20). To ensure that both deletion rules can apply in a single derivation, vowel deletion should apply before consonant deletion.

(20) Deletion Rules

1. Vowel deletion: $V \rightarrow \emptyset / \&C_$ (& indicates a clitic boundary)
2. Consonant deletion: $C \rightarrow \emptyset / \&_$ (& indicates a clitic boundary)

Next, the probability percentage for the application of each deletion rule must be established. A model for calculating this probability can be constructed based on the frequency of the word that the rule applies to relative to other items in the lexicon. The model also will have to account for the fact that vowel deletion was less common than consonant deletion. The construction of such a model, however, is beyond the scope of this paper. Thus, for the sake of illustration of this analysis, the probability percentages will be based on the deletion percentages observed in the experiment. Consonant deletion occurred in 40% of productions with high frequency hosts and in 5% of productions with low frequency hosts. Vowel deletion occurred in 5% of productions with high frequency hosts and in 2% of productions with low frequency hosts⁶. The grammar will consist of rules, labeled [+optional] and a model that assigns a probability percentage for the application of each rule. Examples of evaluation grammars for high and low frequency hosts are presented in (21) below.

(21) Grammars

High frequency hosts:

1. Vowel deletion: $V \rightarrow \emptyset / \&C_$ [+optional, **5%**]
2. Consonant deletion: $C \rightarrow \emptyset / \&_$ [+optional, **40%**]

Low frequency hosts:

1. Vowel deletion: $V \rightarrow \emptyset / \&C_$ [+optional, **2%**]
2. Consonant deletion: $C \rightarrow \emptyset / \&_$ [+optional, **5%**]

A sample derivation of the four possible surface forms for the underlying form *natanleɣa* is presented in (22) below.

⁵ For example, the inflected preposition /lahem/ ‘to them’ is commonly produced as [laem].

⁶ The numbers are rounded for simplification.

(22) Derivations

a)	/natan&lexa/
V-deletion, 5%	–
C-deletion, 40%	–
	[natanlexa]

b)	/natan&lexa/
V-deletion, 5%	natanlxa
C-deletion, 40%	–
	[natanlxa]

c)	/natan&lexa/
V-deletion, 5%	–
C-deletion, 40%	natanexa
	[natanexa]

d)	/natan&lexa/
V-deletion, 5%	natanlxa
C-deletion, 40%	natanxa
	[natanxa]

In (22a) neither rule applies, resulting in a surface form with no deletion. In (22b) only vowel deletion applies, while in (22c) only consonant deletion applies. In (22d) both deletion rules apply, resulting in a surface form with deletion of both segments. Thus, the rule-based theory can generate the four observed surface forms of the clitic. A model that predicts the likelihood of application of the deletion rules, however, is yet to be constructed.

8. Conclusion

This paper had two goals. The first goal was to determine whether hosts with low token frequency block the cliticization of function words onto them. It was found that cliticization occurs regardless of the token frequency of the potential host. This is unlike some other phonological processes that affect high-frequency words more than they affect low-frequency words (e.g., final *-t/d* deletion in English (Coetzee (2008))). It may be the case that token frequency only plays a role in segmental processes and does not affect processes that are purely prosodic. While some segmental processes have the potential of altering the prosodic structure of a word by changing the number of syllables or syllable weight, prosodic processes do not directly alter the segmental structure of a word.

The second goal of this paper was to test whether the token frequency of a host plays a role in the production of the clitic itself. It was found that clitics are produced in their reduced form significantly more often when paired with high-frequency hosts. This can indicate that cliticization itself is not an absolute phenomenon but rather a gradient one.

Finally, it was shown that both Optimality Theory and the rule-based theory can provide a correct account of the findings. In OT, the Lexical Distribution Function approach developed in Coetzee (2008) successfully predicts the likelihood of segmental reduction in the clitic based on the frequency of the host. In the rule-based theory, a model that makes such predictions remains to be developed.

Several additional issues that were discussed in this paper require future research. Token frequency was found to play a role in the segmental realization of the clitic but not in the prosodic process of cliticization. Further research is required to test the role of token frequency in processes that are purely prosodic.

Furthermore, the current research only examined the edges of the frequency spectrum. To determine whether cliticization is indeed gradient, future research should examine the production patterns of the clitic with hosts from the entire frequency spectrum.

Additional evidence for a non-absolute approach to cliticization can come from examining perception. This can be done by comparing the phonological processing of reduced clitics paired with hosts that vary in token frequency. For high-frequency hosts, a reduced form of a clitic is expected to be more natural and evoke less processing difficulty than for low-frequency hosts. I will attempt to analyze the perception of such host-reduced clitic combinations in future research.

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⁷ This paper is a revised version of: Selkirk, Elisabeth. 1996. The prosodic structure of function words. In J. L. Morgan & K. Demuth (Eds.), *Signal to syntax: Bootstrapping from speech to grammar in early acquisition* (pp. 187-213). Hillsdale, NJ, US: Lawrence Erlbaum Associates, Inc.

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Appendices

Appendix A: Experiment materials



תקציר

מחקר זה בוחן את תפקיד השכיחות בדקדוקים פונולוגיים. באופן ספציפי, אני מתמקדת בהשפעות שיש לשכיחות של פעלים על מילות היחס הנטויות שעוברות הצמדה (קליטיסיזציה, cliticization) ומתחברות אליהם.

המחקר מורכב מניסוי הפקה, ולאחריו ניתוח פרורמלי של הממצאים. בניסוי, 27 דוברי עברית שפת אם הפיקו משפטים עם פעלים הנבדלים בשכיחות שלהם אשר שימשו כמארחים (hosts) פוטנציאליים, ואחריהם מילת היחס הנטויה *lexa* 'לך'.

נתוני השכיחות נלקחו מקורפוס ה-Hetenten הזמין באתר Sketch Engine (<https://www.sketchengine.eu>). הפעלים שהופיעו בניסוי היו פעלים בעלי שכיחות גבוהה או שכיחות נמוכה. התוצאות מראות שקליטיסיזציה עשויה להתרחש ללא קשר לשכיחות של המארח. המימוש הסגמנטלי של הקליטיק, לעומת זאת, נמצא מושפע משכיחות המארח - ככל ששכיחות המארח גבוהה יותר, כך הקליטיקס מופחתים יותר מבחינה סגמנטלית.

הממצאים העיקריים של מחקר זה הם כפולים. ראשית, ניתן להראות כי השכיחות הינה חלק אינהרנטי מהדקדוק הפונולוגי. שנית, יש להתייחס לקליטיסיזציה כתופעה הדרגתית - ככל ששכיחות המארח גבוהה יותר, כך הקליטיק יעבור יותר הפחתה סגמנטית. הדבר עשוי להצביע על כך שיש לראות את הפונולוגיה בכללותה כמערכת הדרגתית, ולא כמערכת בינארית או אונארית.



אוניברסיטת תל-אביב

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על ידי

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